

Generative flow model on distance geometry for predicting transition states of chemical reactions

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This paper addresses the challenge of predicting transition states, which are essential for understanding reaction mechanisms but difficult to obtain through experiments or conventional computations. We formulate transition state prediction as conditional probability flow learning in molecular distance geometry space rather than Cartesian coordinates. By operating on interatomic distances, the generative flow model on distance geometry employs a chemically constrained representation that captures bond reorganization. An optimal transport inspired evolution path directs the generative process toward chemically consistent intermediates through a dual branch network architecture. On the benchmark datasets, the proposed model improves pre-optimization structural fidelity over Cartesian baselines, accelerates transition state refinement, and generalizes to unseen reactions. This work shows that aligning generative modeling with chemically constrained geometric representations enables practical and accelerated transition state modeling pipelines.

Transition states (TSs) refer to saddle points on the potential energy surface (PES) connecting reactant and product minima¹. Predicting TS is essential for understanding chemical reactions, as the energy barrier between TS and reactant governs the reaction kinetics, while the geometric configuration of TS offers insights into the reaction mechanism². These understandings serve as the cornerstone for wide applications, including the construction of complex reaction networks³, design of new catalysts^{4,5}, optimization of reaction conditions⁶, the development of novel synthetic routes⁷, etc.

The femtosecond-scale lifetime of TSs makes them difficult to characterize experimentally, and only few studies have directly captured TS structures or properties. For instance, ultrafast electron diffraction has been used to get a direct observation of TS⁸. While its high cost prevents widespread application. Alternatively, TS exploration can be performed systematically using search algorithms combined with quantum mechanical calculations like density functional theory (DFT). Common TS search algorithms include growing string method⁹, nudged elastic band (NEB) method¹⁰, and Hessian-based optimization methods^{11,12}. This methodology has been applied to construct complex

reaction networks^{13,14}. However, such computations are computationally expensive and prone to convergence issues.

The emergence of machine learning has led to efficient methods for predicting TSs, overcoming the limitations of traditional approaches. Existing methods fall into two main paradigms. The first category employs Machine Learning Interatomic Potential (MLIP) to iteratively refine initial configurations of TSs, bypassing the need for expensive DFT calculations. NeuralNEB¹⁵ trains an MLIP for the NEB algorithm to approximate the minimum energy path. MLHessian-TSOpt¹⁶ trains a fully differentiable MLIP NewtonNet, which is then used to predict analytical Hessians for saddle point optimization to find TS. Another mainstream approaches utilize machine learning to predict TS directly. Early works generate TSs in one-step manner. For instance, TS-Gen¹⁷ employs a graph neural network to predict the distance matrix, followed by least-squares optimization to reconstruct the TS coordinates. TSNet¹⁸ proposes an end-to-end network structure to predict TS coordinates directly. However, their experiments are limited to specific reaction types. Beyond these, TS-GAN¹⁹ adopts a generative adversarial network (GAN) to map the reactant-product space and

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generate TS distance matrix. Choi²⁰ proposes a PSI-based model for predicting TSs of general organic reactions, in which the interatomic distances of TS are derived from atomic pair features reflecting reactant, product, and linear interpolation of their Cartesian coordinates. However, these one-shot formulations inherently struggle to capture the complexity of bond evolution in chemical reactions. Recently, multi-step generation methods are also utilized for generating TS, achieving state-of-the-art results. OA-ReactDiff²¹ develops an object-aware SE(3)-equivariant diffusion model that satisfies all physical symmetries and constraints for generating sets of reactant, TS, and product structures. However, an additional confidence model is required to select the best TS from randomly generated samples. To fix this, React-OT²² utilizes an optimal transport approach for generating unique TS from reactant and product.

The essence of TS prediction lies in modeling the evolution of chemical bonds, a process naturally expressed through interatomic distances. While Cartesian coordinates provide a complete spatial representation, they obscure the direct geometric correspondence to bond rearrangements and are prone to unphysical intermediates (e.g., atom clashes) when modeling the continuous generative process. In contrast, distance geometry space intrinsically captures this chemical essence, providing a physically grounded manifold for representing chemical reactions. Meanwhile, state-of-the-art generative models have demonstrated remarkable power in modeling complex molecular distributions. However, their potential remains not fully exploited for TS prediction, as their generative capability is often constrained by a representational mismatch.

In this work, we propose a framework for transition state prediction via optimal transport conditional flow matching in molecular distance geometry space, dubbed TS-DFM. Within this framework, TS generation is formulated as learning a conditional velocity field that transforms an initial distribution toward the target TS distribution on the intrinsic manifold of bond evolution, i.e., the distance geometry space. This principled integration unifies the strong inductive bias of distance geometry with the flexibility of flow-based generative modeling. Specifically, we construct the source distribution via physically informed interpolation in distance geometry space that inspired by IDPP²³. This ensures chemically plausible structures along the flow trajectory. Therefore, the learned velocity field acts as a refinement process that continuously steers the initial geometry toward the true TS. To model this corresponding conditional velocity field, we design TSDVNet, a dedicated dual-branch neural architecture that explicitly fuses reactant and product information to guide the transformation within this constrained space. By aligning generative modeling with the intrinsic geometry of chemical change, TS-DFM establishes a coherent and effective paradigm for TS prediction. On Transition1x dataset²⁴, TS-DFM achieves a 30% higher average structural accuracy than the best-performing baseline React-OT that performs Cartesian-space flow matching. The predicted structures serve as superior initializations for subsequent climbing-image nudged elastic band (CI-NEB) optimization, converging three times faster than from IDPP and at least 10% faster than other TS prediction methods. Moreover, TS-DFM exhibits significantly better generalization ability, outperforming React-OT by at least 16% on unseen reaction types, underscoring the robustness of its physically-consistent formulation. These results suggest that TS-DFM advances the field not only through improved performance but by offering a more insightful and principled framework for generative modeling in chemistry, where fidelity to the underlying physical manifold is key to unlocking the full potential of deep generative models.

Results

Overview of TS-DFM

Current state-of-the-art TS prediction methods have adopted multi-step generative paradigms (e.g., diffusion or flow matching) in Cartesian coordinate space. While this choice is computationally

straightforward, the continuous generative process in Cartesian trajectories often sample unphysical regions like atom clashes and, more critically, obscure the direct geometric evolution of chemical bonds—the essence of chemical reactions. An alternative line of work operates directly in distance geometry space, which provides a more intrinsic representation aligned with bond breaking and forming. However, these distance-based approaches are often limited by suboptimal network architectures or one-step prediction schemes, failing to fully leverage the power and stability of modern continuous generative frameworks. To address this, we propose TS-DFM, a framework that integrates powerful optimal transport conditional flow matching with the natural manifold of chemical change, i.e., the molecular distance geometry space. This principled synthesis ensures that every step of the learned transformation respects geometric constraints, avoiding unphysical intermediates. Consequently, TS-DFM achieves a deeper alignment between generative modeling and the underlying physics of chemical transformations, leading to more accurate and robust TS predictions.

Specifically, our framework begins by mapping the reactant and product structure to the pairwise distance matrices $\mathbf{D}_R$ and $\mathbf{D}_P$. We then formulate a conditional flow matching objective to learn the transformation from an initial distribution toward the target TS, conditioned on $\mathbf{D}_R$, $\mathbf{D}_P$, and atom types $\mathbf{Z}$. Unlike standard flow matching which often starts from Gaussian noise, TS-DFM takes the averaged distance matrix $(\mathbf{D}_R + \mathbf{D}_P)/2$ for initialization, serving as a physically informed start point. This choice circumvents unreasonable bond lengths that arise from simple Cartesian linear interpolation, thereby improving the accuracy of the generative process. After predicting the distance matrix of TS, its atomic coordinates are recovered via nonlinear optimization. To address the fact that distance geometry cannot distinguish enantiomers, the nonlinear optimization is anchored to the known 3D structures of the reactant and product. The initial geometry of the optimization is generated by linear interpolation between reactant and product coordinates, and subsequent nonlinear optimization refines the predicted TS coordinates while explicitly preserving the stereochemical configuration. The details are described in “Methods” section. This framework is illustrated in Fig. 1a, in which we demonstrate the learned velocity field $\mathbf{v}_\theta$ transports the source distribution p_0 denoting the initial guess to the target distribution p_1 representing the TS.

The core of our implementation lies in designing the optimal transport based conditional flow matching model to generate the distance matrix of TS. To implement distance geometry evolution, we designed a network named TSDVNet to model the velocity field that drives the initial distance matrix toward that of TS. As illustrated in Fig. 1b, the network takes $\mathbf{D}_R$, $\mathbf{D}_P$ and $\mathbf{Z}$ as conditional inputs. Their representations, generated via an additional branch, are fused into the main branch to facilitate the prediction of the velocity field. We here just briefly describe the model and network design of TS-DFM, and the details are presented in Details about TS-DFM and Network Structure of TSDVNet in “Methods” section, respectively. In the following sections, we will evaluate TS-DFM by demonstrating its superior prediction accuracy and generalization ability, and exploring its utility in practical workflows to accelerate CI-NEB optimization and discover alternative reaction pathways.

Generating High-quality TS from TS-DFM

We first trained TS-DFM on the Transition1x dataset. The data is generated by performing CI-NEB calculations with DFT at the ω B97x/6-31G(d) level on 10k organic reactions, resulting in 9.6 million calculated energies and forces for molecular configurations on and around the reaction pathways. The reactants, products and TSs are extracted from the converged paths of CI-NEB calculations. In our experiments, we use the default data split provided by the officially released

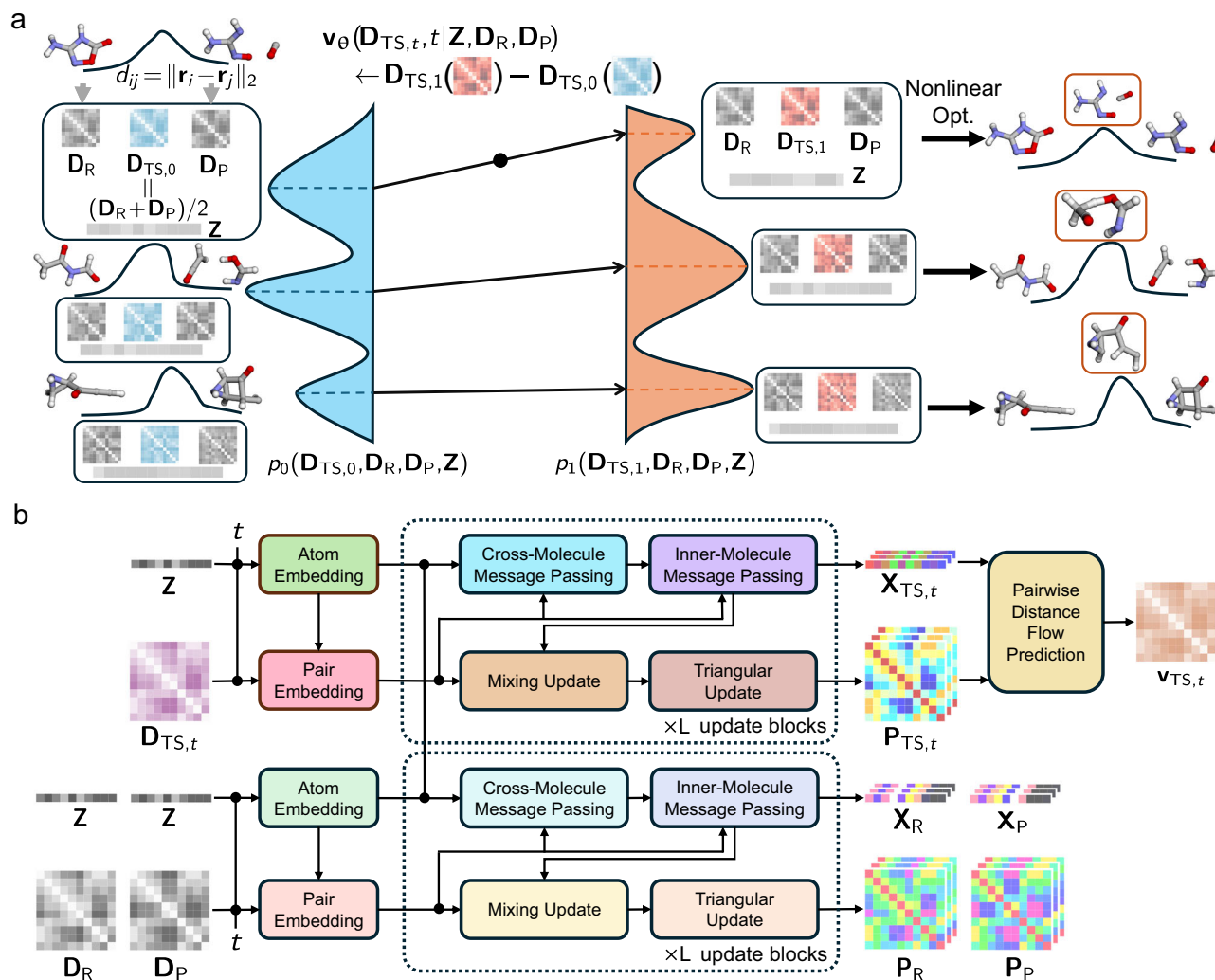


Fig. 1 | Overview of TS-DFM and TSDVNet. a Schematic representations of TS-DFM. Let p_0 and p_1 denote the data distributions over the initial guessed and true TSs of chemical reactions, respectively. Initial guessed and true TSs from the same reaction are coupled. We learn a linear velocity field in the distance geometry space $\mathbf{v}_\theta$ to evolve an initial guessed TS $\mathbf{D}_{TS,0}$ toward the corresponding true TS $\mathbf{D}_{TS,1}$.

b The network structure TSDVNet for learning the velocity field. The network consists of two branches with identical architecture but independent parameters. The upper branch predicts the velocity field, while the lower branch encodes the representations of reactant and product. Features from both branches are integrated by feature fusion operations to produce enriched representations.

database. This data split exhibits no overlap in chemical composition, constituting a stringent and realistic evaluation protocol.

To validate the effectiveness of TS-DFM, we compare it with the following four representative baseline approaches. NeuralNEB¹⁵ utilizes a MLIP as a surrogate PES for NEB calculation. In NEB calculations, we adopt the settings in ref. 15, utilizing IDPP for initialization and the climbing-image method for iterative optimization. The details of MLIP training and NEB configurations are elaborated in Implementation Details of Methods section. PSI-based model²⁰ has the best performance among one-step generation methods. OA-ReactDiff²¹ is a representative approach that generates TSs randomly. For comparison, we follow^{21,22} to sample 20 structures for each test reaction and keep the one with the lowest RMSD. React-OT²² achieves SOTA results among the existing TS prediction methods. All baseline methods are re-trained and re-evaluated using their officially released codes under the same data split to ensure fair comparison.

We compare the performance of different approaches on evaluation metrics that fall into two categories: structural error and energetic properties. The structural error, measured by root mean squared distance (RMSD) and distance mean absolute error (DMAE), assesses the geometric deviation of the predicted TS. The second category evaluates the potential energy properties derived from DFT

calculations on the predicted TS. This includes the absolute error in TS energy $|\Delta E_{TS}|$, the maximum and root-mean-square of atomic forces, i.e., $f_{\max}$ and F_{rms} . As a saddle point, a valid TS must exhibit near-zero forces. Additionally, harmonic analysis is performed based on the Hessian matrix, and the predicted structure with only one imaginary frequency is identified as a saddle point. The detailed description and mathematical definitions of these metrics are in Evaluation Metrics of the Methods section.

In comparison with baseline methods, TS-DFM achieves the best overall performance. It demonstrates superior structural accuracy, with a mean RMSD of 0.1828 Å and a mean DMAE of 0.0607 Å, outperforming the best baseline React-OT by approximately 30% on both metrics. Notably, TS-DFM achieves this with only half model parameters (5.2M versus 10.6M). TS-DFM also has the lowest average $|\Delta E_{TS}|$ of 0.1614 eV, and 65.5% of its predicted TSs are located at saddle points, both of which surpass the second-best method NeuralNEB (0.2970 eV and 63.41%, respectively). By iteratively refining image configurations toward saddle points in NEB calculations, NeuralNEB achieves the best performance in force-related metrics ($f_{\max}$: 1.1190 eV/Å; F_{rms} : 0.5588 eV/Å). TS-DFM performs most closely to it ($f_{\max}$: 1.6482 eV/Å; F_{rms} : 0.8227 eV/Å), significantly outperforming the other three methods. In terms of efficiency, TS-DFM generates TS in only 1.2 s

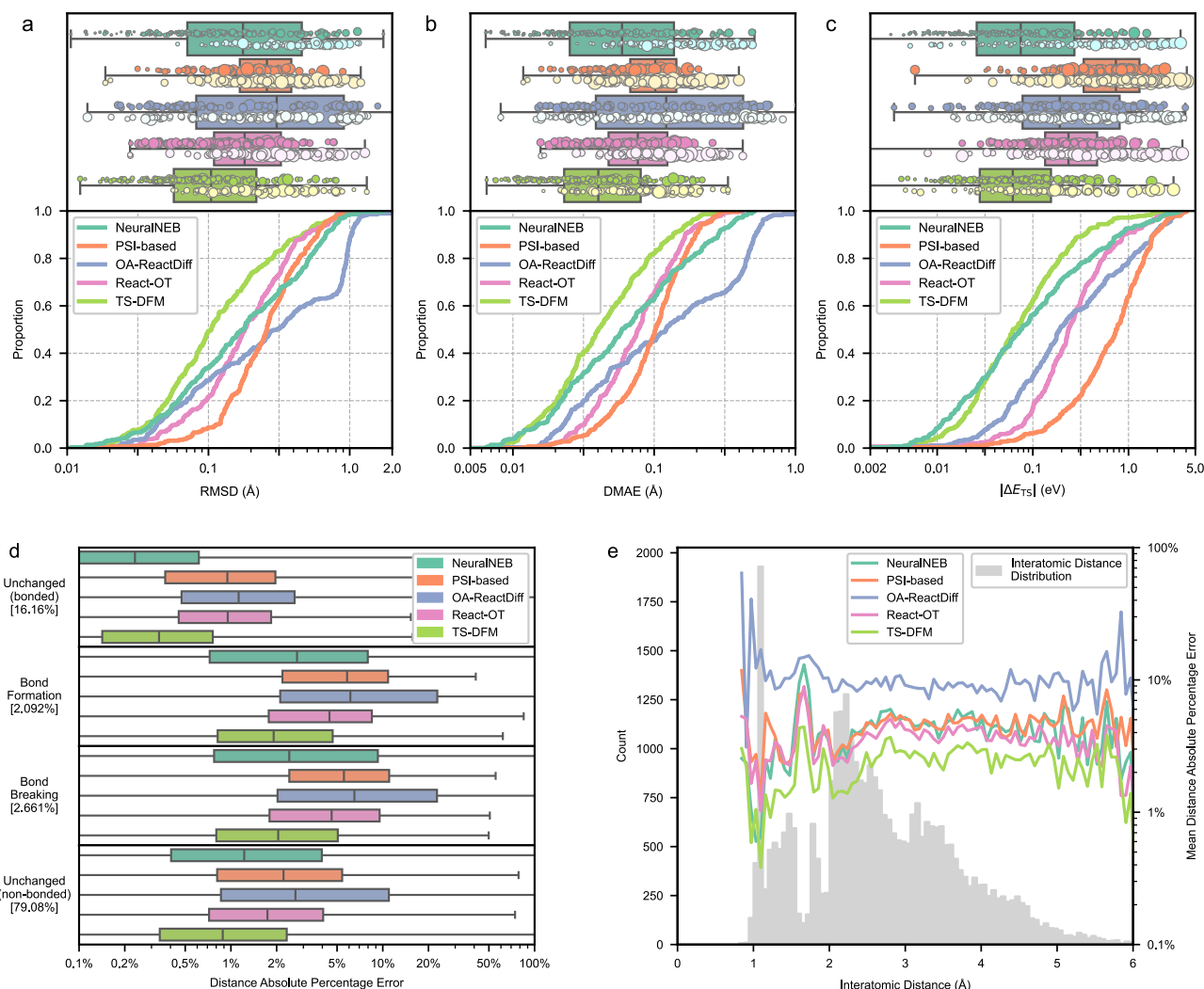


Fig. 2 | Error analysis of different methods. **a–c** The boxplots (upper) and cumulative probabilities (lower) for RMSD, DMAE and $|\Delta E_{TS}|$ of the predicted TSs on test dataset, respectively. The prediction error for each test data is also plotted as a point on the boxplot, where the point's size is proportional to its $f_{\max}$ value, its color (dark or light) denotes whether the predicted TS lies at a saddle point or not. **d** Boxplots for the absolute percentage errors in pairwise distances of predicted

TSs, categorized by the type of bond change during the chemical reaction. Percentages in brackets indicate the proportion of each bond change type within the entire test set. **e** Average absolute percentage errors and numbers of atom pairs w.r.t. pairwise distances. The gray histogram shows the distribution of interatomic distances. The colored line represents the mean absolute percentage error computed over different distance intervals.

on average, including interatomic distance prediction and subsequent nonlinear coordinate reconstruction. This time cost is three times slower than React-OT but 50 times faster than NeuralNEB, achieving a practical trade-off between speed and accuracy. It is worth noting that the official React-OT results were reported under a different data splitting and evaluation approaches. For comprehensive comparison, we also evaluate TS-DFM under React-OT's protocol (see Supplementary Note 3). We further conduct two ablation studies to investigate the source of TS-DFM's superior performance (see Supplementary Tables 1 and 2). The results demonstrate the effectiveness of TSDVNet, linear interpolation in pairwise distance space, and flow matching objective all contribute to the superior performance of TS-DFM.

Figure 2a–c illustrates the distribution of different evaluation metrics in the test set. TS-DFM achieves the best cumulative probabilities of RMSD and DMAE among the compared methods. For $|\Delta E_{TS}|$, NeuralNEB performs the best at values below approximately the 40th percentile of the error distribution, while TS-DFM surpasses other methods at values above this threshold. To determine whether TS-DFM better captures the changes in interatomic distances, we further compare the distributions of absolute percentage errors in the interatomic distances

of predicted TSs across different methods, as shown in Fig. 2d, e. In Fig. 2d, we apply the method proposed in ref. 25 to infer chemical bonds in reactants and products. Atom pairs are classified into four categories based on their bonding evolution during the reactions. Among five approaches, NeuralNEB performs best in predicting distances of unchanged chemical bonds, while TS-DFM achieves the highest accuracy for the other three bond-change types. Notably, for bond breaking and formation—the key processes in chemical reactions, TS-DFM slightly outperforms NeuralNEB and shows at least 2% improvement over the other three methods in the median values. We also analyze the average distance percentage error as a function of interatomic distance, with results presented in Fig. 2e. These results demonstrate that TS-DFM performs the best across almost all interatomic distances. The weakest performance occurs at approximately 1.6 Å, a distance that corresponds to the typical range of chemical bond breaking and formation. The significant errors in this region can likely be attributed to data scarcity, as this is commonly observed in other data-limited regions. In conclusion, TS-DFM more effectively captures the changes in interatomic distances during chemical reactions compared to previous methods, leading to superior accuracy in the predicted TSs.

TS-DFM provides better initialization for TS search algorithms

As illustrated in the previous section, the predicted TSs from TS-DFM still have gaps to the saddle points. Although not perfectly accurate, the predicted structures offer useful initial configurations that enhance convergence in TS search algorithms. In this section, we take NEB calculation as an example to evaluate their suitability as starting configurations. Due to the high computational cost of DFT, MLIP is used as a surrogate PES. The traditional IDPP interpolation is set as a baseline initialization method. For PSI-based model, OA-ReactDiff, React-OT, and our proposed TS-DFM, the predicted TS from each method serves as an additional anchor point in NEB images between reactant and product. Intermediate images are then placed along reactant-TS and TS-product segments, followed by IDPP initialization.

To comprehensively evaluate the performance of different approaches, Tab. 1 presents a comparative analysis of both directly predicted and CI-NEB optimized TSs. The upper part summarizes the results of direct prediction methods mentioned in the previous section, while the lower part demonstrates the performance after CI-NEB optimization. This integrated comparison allows a clear assessment of how initialization methods influence the quality of optimized TSs. In comparison with the results from upper and lower parts of Table 1, the optimized TSs from PSI-based model, OA-ReactDiff, and React-OT demonstrate significant refinements upon their directly predicted structures. However, OA-ReactDiff suffers from severe convergence problem in subsequent NEB calculations, where only 59.23% of test data successfully converged in NEB calculations and exhibits the worst average performance. The rest of the approaches all surpass IDPP, and our proposed TS-DFM achieves the best results across all metrics. For TS-DFM, the structure refinements are mainly represented in the decrease of $f_{\max}$ and F_{rms} , with only a slight change in RMSD, DMAE, and $|\Delta E_{\text{TS}}|$. Similar to Fig. 2, the distributions and cumulative probabilities of different metrics after NEB optimization are illustrated in Supplementary Fig. 2. TS-DFM achieves the best cumulative probabilities on three metrics among the compared initialization methods. Additionally, TS-DFM has the best convergence time on average, outperforming other initialization methods by at least 10%. Therefore, the predicted TS of TS-DFM needs less iterations in NEB calculation than the compared methods. The longer inference time of TS-DFM is compensated by its faster NEB convergence, making it the most efficient method overall.

To consolidate our conclusions via higher-fidelity calculations, we further perform DFT-based (at $\omega\text{B97x/6-31G(d)}$ level of theory) CI-NEB optimizations on a subset of 50 test cases selected via stratified sampling to ensure chemical diversity. Given the computational expense, we compare our proposed TS-DFM only with the best-performing baseline, React-OT. The results are detailed in Supplementary Tab. 6. They confirm that TS-DFM achieves higher structural accuracy for the converged TSs, which aligns with the conclusions from MLIP-based calculations. Furthermore, TS-DFM provides a critical advantage in computational efficiency. Using the predictions of TS-DFM as additional anchors in NEB initialization leads to significantly faster convergence, saving approximately 5 h of DFT computational time per reaction on average. This efficiency gain stems from TS-DFM providing a starting geometry closer to the true saddle point, thereby reducing the number of expensive force and energy evaluations required during CI-NEB optimization. This reduction in convergence time is of substantial practical importance. For instance, in computational catalysis and reaction mechanism studies, where screening hundreds of reactions is common, the computational efficiency directly translates to lower resource costs and accelerated research cycles.

Discovering alternative reaction pathways

Chemical reactions with multiple TSs are frequently observed in polyatomic systems, where a single pair of reactant and product minima can be connected by more than one first-order saddle point

Table 1 | Statistics of evaluation metrics for the predicted TSs from different approaches and CI-NEB optimized TSs from different initialization methods

Method Type	Approaches/ Init. method	Params	Time (s)	% of NEB convergence	% of saddle pts	RMSD (Å)		DMAE (Å)		$ \Delta E_{\text{TS}} $ (eV)		$f_{\max}$ (eV/Å)		F_{rms} (eV/Å)	
						mean	median	mean	median	mean	median	mean	median	mean	median
Direct Prediction	PSI-based	11.9M	0.1	-	28.57%	0.3035	0.2594	0.1126	0.1019	0.9191	0.7383	4.4587	3.8852	2.1864	1.9574
	OA-ReactDiff	10.6M	5.1	-	40.42%	0.4845	0.3043	0.2259	0.1217	0.5979	0.1912	2.6850	2.3206	1.3356	1.1764
	React-OT	10.6M	0.4	-	55.75%	0.2436	0.1803	0.0934	0.0766	0.4272	0.2359	3.4365	2.9620	1.7010	1.5123
	TS-DFM	5.2M	1.2	-	65.51%	0.1828	0.1000	0.0607	0.0402	0.1614	0.0617	1.6482	1.2259	0.8227	0.6100
CI-NEB Optimized	IDPP(NeuralNEB)	5.0M	70.0	88.50%	63.41%	0.2836	0.1756	0.1066	0.0592	0.2970	0.0748	1.1190	0.7018	0.5588	0.3598
	PSI-based	5.0M+11.9M	33.9	86.41%	67.25%	0.2332	0.1371	0.0835	0.0512	0.2076	0.0645	1.0206	0.6250	0.5177	0.3293
	OA-ReactDiff	5.0M +10.6M	28.4	59.23%	60.98%	0.2944	0.1588	0.1340	0.0562	0.6885	0.0717	1.4550	0.7592	0.7353	0.4019
	React-OT	5.0M +10.6M	22.9	90.59%	70.38%	0.2049	0.1091	0.0743	0.0423	0.1712	0.0499	0.9803	0.6153	0.4851	0.3080
	TS-DFM	5.0M+5.2M	19.7	91.29%	74.22%	0.1832	0.0862	0.0693	0.0352	0.1413	0.0444	0.9166	0.6014	0.4607	0.3024

that corresponds to different reaction mechanisms or conformational pathways^{26,27}. Identifying the TS with the lowest energy barrier is challenging and typically requires the exploration of multiple TS structures. In this section, we focus on exploring the ability of TS-DFM as a component of a workflow for generating multiple TSs, which is useful for discovering alternative reaction pathways. Since TS-DFM generates TSs deterministically, one approach to obtaining diverse TSs is to introduce random perturbations to the reactants and products, as demonstrated in ref. 20. To ensure these perturbed structures remain physically reasonable, we utilize normal mode sampling (NMS)²⁸ with intentionally limited amplitude for both reactants and products (See Implementation Details in Methods section). This constrained sampling keeps the perturbed structures within the same local minima as the originals, thereby focusing the search on alternative TSs for the same paired reactant and product. The perturbed reactant and product are then fed into TS-DFM. Shifting the input distribution alters the target distribution of the learned flow, enabling efficient generation of diverse TS candidates that correspond to different reaction pathways while still being intended to connect the original, unperturbed endpoints. However, the injected noise can compromise the accuracy of the generated TS geometries. To obtain precise saddle points, we refine each candidate with a CI-NEB calculation. For each candidate, we construct an initial chain by inserting the TS-DFM output as an additional anchor point between the unperturbed reactant and product endpoints. During band relaxation, the highest-energy image is driven toward the saddle point while the chain of states keeps it naturally coupled to both endpoints. To enable rapid screening of alternative pathways, these CI-NEB calculations are performed using a MLIP as a surrogate PES. The full pipeline can generate and refine a TS candidate in about one minute, making it practical for the rapid exploration of diverse reaction channels.

We investigate three different reactions from the test set of Transition1x as examples, as illustrated in Fig. 3a. Molecular structures are visualized using py3DMol²⁹. For each reaction, we collect 100 different TSs that converged in CI-NEB calculation. To visualize the differences of these collected TSs, they are further represented by the Coulomb matrix³⁰ followed by principal component analysis (PCA). Figure 3b, c present PCA plots using the first two PCA dimensions of the TS representations before and after CI-NEB optimization, respectively. In panel c, the optimized structures are clustered using the K-means algorithm applied in the PCA space. To trace the origin of these distinct TSs, the points in panel b are colored according to the cluster assignments of their corresponding optimized structures in panel c. The results reveal that the three reactions exhibit different numbers of distinct clusters in panel c, with each cluster corresponding to a unique, stable TS geometry. For simplicity, we denote the TS corresponds to cluster *i* as TS-*i*. Then, we plot the RMSD, DMAE, and $|\Delta E_{TS}|$ w.r.t. the category of cluster, whose results are shown in Fig. 3d–f, respectively. For the leftmost reaction in Fig. 3, only one TS is found, whose structure is close to the reference TS. Two TSs with close energy barriers are found for the reaction in the middle of Fig. 3. In the rightmost case of Fig. 3, TS-1 is closest to the reference. Interestingly, TS-0 and TS-2 correspond to new TSs with different reaction mechanisms from the reference. Moreover, the statistics of $|\Delta E_{TS}|$ for TS-2 outperform TS-1, indicating that a TS with lower energy barrier might be found.

To validate the predicted TSs, intrinsic reaction coordinate (IRC) calculations are carried out using DFT at ω B97x/6-31G(d) level of theory for the lowest-energy structure within each cluster. Due to the inaccuracy of MLIP, these structures are first refined via DFT-based saddle point optimization before executing IRC calculations. The details of these calculations are in the Implementation Details of Methods section. It is confirmed that all TSs shown in Fig. 3c could obtain the intended reactant and product molecules in IRC calculations (see Supplementary Fig. 3). Specifically, for the cases in the

middle of Fig. 3, the refined structures of TS-1 has potential energy close to that of TS-0. Compared to the reference structure, the reactant geometry derived from IRC calculation of TS-1 is a distinct conformer of the same molecule. In the rightmost case of Fig. 3, the optimized structure of TS-1 is almost identical to the reference TS. However, the optimized structure of TS-0 is close to that of TS-2. If MLIP is used in IRC calculations instead, TS-0 is located at the saddle point and the intended reactant and product structures are obtained, indicating that TS-0 is an artifact induced by the inaccuracy of MLIP. Then, it is confirmed that the reference reactant and product molecules could be reached in IRC calculations using the optimized structure of TS-1 and TS-2. In addition, the potential energy of the optimized TS-2 is 0.47 eV lower than that of TS-1, demonstrating that a better reaction path is found. To further validate our findings regarding MLIP inaccuracies, we conduct an additional experiment for the rightmost case in Fig. 3. Ten samples are randomly selected from each of the three clusters identified in panel c, yielding 30 samples in total. Their corresponding initial structures produced by TS-DFM, i.e., the geometries prior to any MLIP-based CI-NEB optimization, are then used as starting points for DFT-based CI-NEB calculations. The results (see Suppl. Fig. 4) confirm that only two clusters are identified in PCA space. The representative structures of cluster 0 and 1 in Supplementary Fig. 4 correspond to cluster 2 and 1 in the third case of Fig. 3, respectively. This confirms that TS-0 is indeed an artifact induced by the inaccuracy of MLIP, and also confirms the validity of the other two TSs. While DFT-based CI-NEB yields more reliable results, its high computational cost makes it impractical for rapid screening of alternative TSs. In contrast, a more feasible and computationally efficient workflow is to first perform fast screening with MLIP-based CI-NEB, followed by selective DFT-based verification.

Currently, we limit the amplitude of NMS perturbations so that the sampled reactant and product structures tend to remain within the same local minima as the original ones. This setting allows the workflow to efficiently explore alternative TSs for a given paired reactant and product. For larger or highly flexible systems, conformational sampling is often utilized to explore diverse reaction mechanisms arising from conformational change^{31,32}. We envision a two-stage pipeline in which extensive conformational sampling of reactants and products is first performed, then the generated conformers can be fed into TS-DFM, optionally paired with the NMS-based perturbations, to generate diverse TS candidates. These candidates can be further refined by either the CI-NEB optimization or alternative single-ended saddle-point optimization methods. Such a combined protocol would extend the applicability of TS-DFM to complex systems.

Superior generalization ability to unseen reactions

To explore the capability of TS-DFM, we conduct additional experiments using the RGD1 dataset³³, which contains 176k organic reactions at the B3LYP-D3/TZVP level and offers a wider variety of reaction types and molecular structures than Transition1x. The data is partitioned into five subsets: Train-id (20k), Valid-id (1k), Test-id (1k), Test-ood-type (1k), and Test-ood-size (1k). The splitting procedure is illustrated in Supplementary Fig. 5. In comparison with Test-id subset, Test-ood-size contains reaction data with larger molecular size, while Test-ood-type contains reaction data with different reaction types. Therefore, this dataset can be used not only to evaluate model performance on complex chemical reactions, but also to assess generalization to unseen molecular structures and reaction types.

Table 2 presents a comparative analysis between TS-DFM and React-OT using our partitioned subset of RGD1. The results demonstrate statistically significant superiority of TS-DFM across three test subsets. Specifically, TS-DFM achieves lower mean RMSD and DMAE values, with at least 16.0% reductions. This superiority is quantified by the positive $\Delta\epsilon$ values, indicating that TS-DFM obtains lower errors than React-OT in a majority of test cases (68.7–86.2%). The consistency

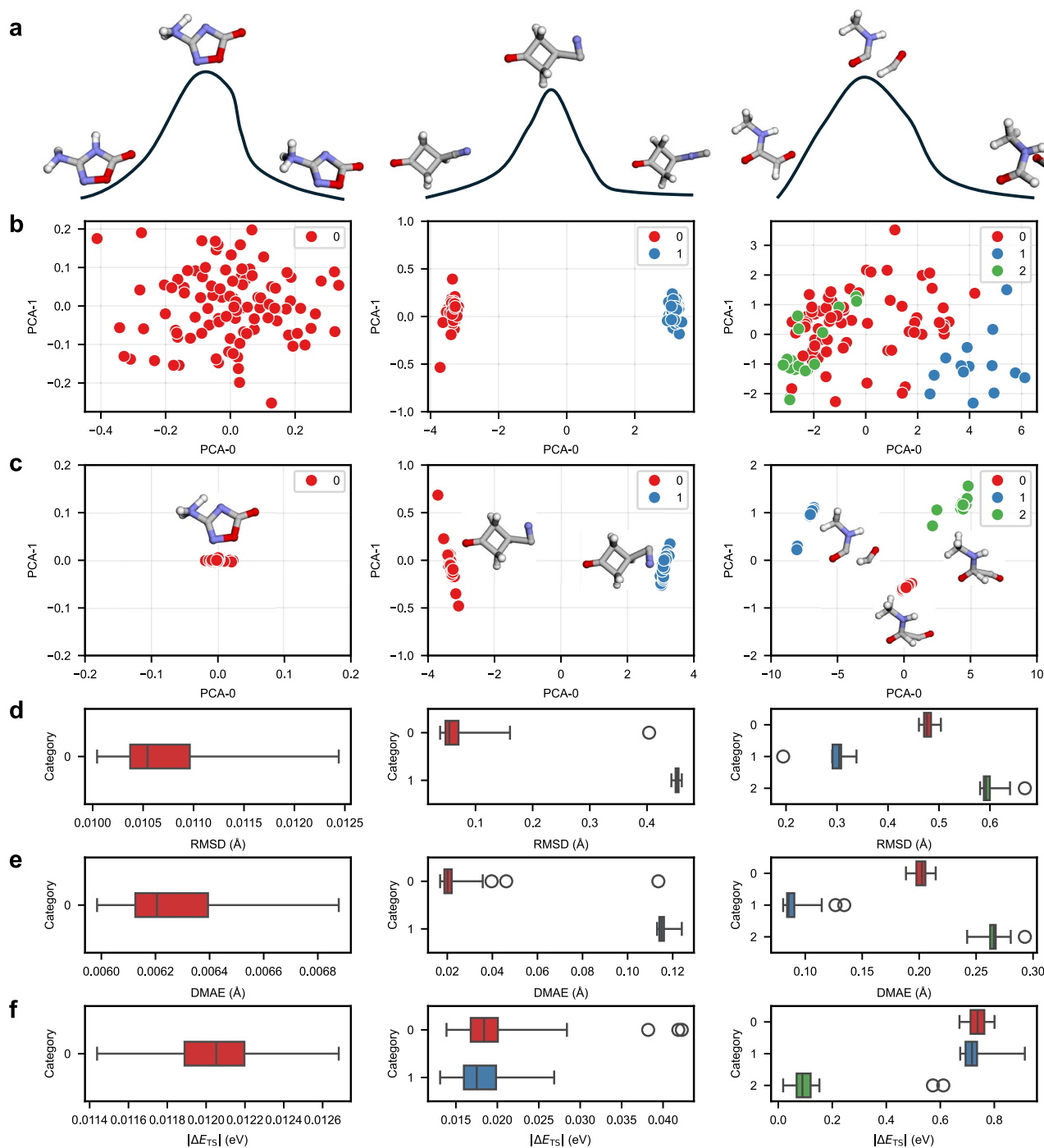


Fig. 3 | Analysis of samples generated in TS exploration. **a** Illustration of three chemical reactions in the test set. The H, C, N and O atoms are colored as white, gray, blue and red, respectively. **b, c** Principal component analysis (PCA) on the Coulomb matrices of 100 generated TSs before and after CI-NEB optimization, respectively. In **c**, the structures are clustered via K-means in the PCA space. For

visual traceability, each point in **(b)** is colored according to the cluster assignment of its corresponding optimized structure in **(c)**. For each cluster, the structure with the lowest potential energy is illustrated nearby. **d–f** The boxplots of RMSD, DMAE, and $|\Delta E_{TS}|$ between the generated samples and referenced TS for each cluster, respectively.

of this advantage is further validated by extremely low p -values ($p \leq 8.51 \times 10^{-44}$) in the Wilcoxon two-sided signed-rank tests, firmly rejecting the null hypothesis of comparable performance. We also conduct bond analysis for the predicted TSs similar to that in Fig. 2d, e, which are presented in the Supplementary Figs. 6 and 7. The results demonstrate that TS-DFM surpasses React-OT in different bond change types and in almost all interatomic distances.

A comparative demonstration for React-OT versus TS-DFM on different subsets is represented in Fig. 4a–f. Each data point represents

one test data, with its x-axis and y-axis values corresponding to the RMSD (or DMAE) of React-OT and TS-DFM, respectively. Generally, the majority of data points in Fig. 4a–f are located under black dashed lines, which is in accordance with the results of Table 2. We notice that there are several test data whose prediction accuracy of TS-DFM and React-OT differ greatly. Figure 4g presents three representative reactions that TS-DFM underperform React-OT, illustrating typical failure modes that lead to large structural errors. The TSs predicted by TS-DFM in these cases suffer from both geometrical inaccuracies (in

Table 2 | Statistics of TS-DFM versus react-OT on RGD1 dataset

Data split	Metrics	React-OT		TS-DFM		Statistics of $\Delta\epsilon^a$			Wilcoxon signed-rank test	
		mean	median	mean	median	% of $\Delta\epsilon > 0$	$\Delta\epsilon_{\text{mean}}$	$\Delta\epsilon_{\text{std}}$	p-value (two-sided)	Rank biserial correlation
Test-id	RMSD (Å)	0.5990	0.5468	0.4388	0.3509	75.0%	0.1602	0.1532	1.33×10^{-65}	0.6245
	DMAE (Å)	0.2191	0.2004	0.1439	0.1146	82.9%	0.0752	0.0730	5.86×10^{-101}	0.7787
Test-ood-size	RMSD (Å)	0.6609	0.6001	0.4913	0.3850	75.4%	0.1696	0.1606	1.75×10^{-65}	0.6239
	DMAE (Å)	0.2479	0.2326	0.1606	0.1329	86.2%	0.0874	0.0862	1.12×10^{-116}	0.8382
Test-ood-type	RMSD (Å)	0.7991	0.7490	0.6692	0.6176	68.7%	0.1298	0.1341	8.51×10^{-44}	0.5067
	DMAE (Å)	0.3491	0.3352	0.2494	0.2199	79.7%	0.0996	0.0951	9.56×10^{-91}	0.7375

^a $\Delta\epsilon = \epsilon_{\text{React-OT}} - \epsilon_{\text{TS-DFM}}$, where ϵ denotes RMSD or DMAE of different approaches.

torsional angles, and fragment positioning) and erroneous bond connections. Nevertheless, TS-DFM captures partially correct reaction mechanisms even in several extreme cases, such as identifying correct bond changes in the first two cases of Fig. 4g. In practice, these could provide reasonable initial guesses for subsequent structure optimization. More test cases are provided in Supplementary Figs. 8, 9, and 10. It should be emphasized that these cases are not indicative of the overall performance, as TS-DFM maintains a higher average accuracy than React-OT.

Statistical analysis of prediction failures

In this section, we statistically analyze prediction failures of TS-DFM, focusing on the relationship between molecular features and prediction errors on Transition1x and RGD1 datasets. To ensure comprehensiveness, the three test subsets of RGD1 dataset are merged during analysis, providing a more diverse distribution of features and errors. The same analysis is also conducted on the best-performing baseline React-OT for comparison. Figure 5 presents a heatmap of correlations between eight molecular features and six prediction errors. Detailed definition and calculation of these features and errors, the correlation analysis methods, and additional visualization of these feature and errors are provided in the Supplement Information.

Overall, the results indicate that reaction complexity is strongly associated with prediction errors for both methods and datasets. On both datasets, the number of bond breaking/formation and the number of reaction center exhibit strong positive correlations with bond breaking/formation errors, angle error, and improper angle error. These features directly quantify the complexity of chemical reactions, making accurate TS prediction more challenging. On RGD1 dataset, a strong correlation is also observed between the number of TS fragments and prediction errors, a relationship that is weaker on the simpler Transition1x dataset. In contrast, the location of the reaction relative to ring structures shows weak correlation with errors across datasets, indicating its minimal impact on prediction accuracy.

Although TS-DFM generally performs better than React-OT, both approaches struggle with complex reactions, highlighting the fundamental challenges in accurately modeling diverse chemical transformations. We propose that TS-DFM could be integrated into a closed-loop workflow to accelerate the construction of complex reaction databases while simultaneously improving model performance: by generating TS candidates for unexplored complex reactions, validating them through high-accuracy quantum chemical calculations, and retraining the model on the augmented dataset. We leave this direction for future work.

Discussion

In this study, we propose TS-DFM, a distance-geometry-based flow matching framework for generating TSs with provided reactant and product structures. Operating in distance geometry space enables effective capture of the fundamental nature of chemical reactions, i.e., the evolution of interatomic distances. Consequently, TS-DFM not only achieves higher accuracy in predicting TSs than SOTA methods,

but also demonstrates superior generalization ability on unseen reactions. Even in some of the worst cases of TS-DFM, the predicted TS can still demonstrate partially correct reaction mechanisms.

Furthermore, TS-DFM holds great potential for integration into diverse workflows across various applications. For example, the TSs predicted by TS-DFM provide better initial structures for TS search algorithms. In our experiments, they are used as initial configurations for CI-NEB optimization, improving both the convergence speed and the accuracy of converged structures. When quantum chemical accuracy is required for reaction pathway analysis, TS-DFM could offer substantial computational savings. As a proof of concept, we also demonstrate the application of TS-DFM in a workflow designed to discover alternative reaction pathways: by employing normal mode sampling to generate thermally activated reactants and products, followed by TS candidates generation via TS-DFM and MLIP-based CI-NEB optimization, multiple TSs with different reaction mechanisms are identified in some chemical reactions. Notably, in one specific case, we identify a reaction pathway that is more favorable than the reference in the dataset. Such workflows hold significant promise for elucidating competitive reaction mechanisms, expanding reaction databases, and accelerating reaction discovery.

Currently, TS-DFM is primarily developed and evaluated for predicting TSs of uncatalyzed, gas-phase organic reactions involving CHON elements. Its applicability to reactions containing heavier or less common elements, to systems involving explicit solvent or aqueous-phase environments, or to catalytic processes has not yet been systematically explored. These domains introduce additional physical interactions like solvation effects, ionic environments, and explicit catalyst-substrate interactions, which are not explicitly represented in the current training data. Nevertheless, our proposed framework is extensible in principle. For instance, modeling catalytic reactions could be approached by explicitly incorporating catalyst fragments into the molecular graph representation. Solvation and aqueous-phase effects could be integrated by augmenting the model with implicit solvation descriptors or by including explicit solvent molecules in the input molecular graphs. Such extensions require both methodological adaptations and expansion of training data beyond current gas-phase reaction datasets. Beyond organic chemistry, applying TS-DFM to biological systems (e.g., enzymatic catalysis or protein conformational changes) and materials science (e.g., phase transition or solid-state diffusion) represents a promising yet challenging research direction. These areas involve larger system sizes and different bonding regimes, which would require scaling the model and retraining on domain-specific data.

Methods

This section introduces the general framework, network design, evaluation metrics, and some implementation details of our proposed TS-DFM. The following content of this section can be summarized as follows.

- (1) We elaborate on the theoretical foundations of flow matching as a generative model, and the formulation of TS-DFM for TS prediction task through a probabilistic generative modeling

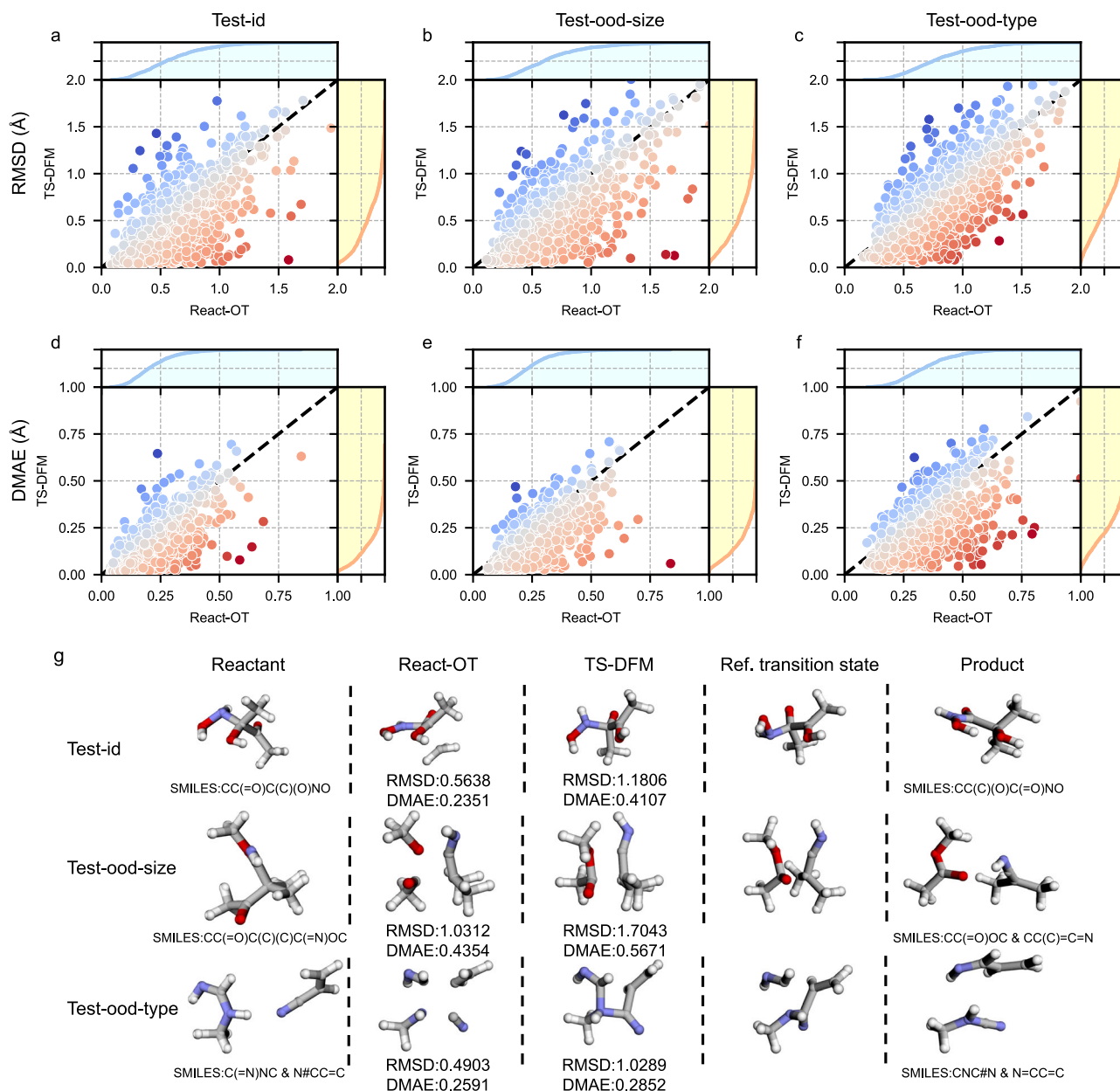


Fig. 4 | Performance comparison and case studies of TS-DFM versus React-OT on different test subsets. **a–c** The RMSD of TS-DFM versus React-OT for each data on Test-id, Test-ood-type and Test-ood-size subsets, respectively. **d–f** The corresponding comparisons for DMAE. The x and y axis correspond to the RMSD or DMAE of React-OT and TS-DFM. On top and right of each subfigure are the respective cumulative probability distributions of React-OT and TS-DFM. A blue-white-red color gradient is used to represent the difference in RMSD or DMAE between the two methods, where darker blue indicates that React-OT performs better, darker red indicates that TS-DFM performs better, and white suggests that the metrics of the two methods are close. **g** Examples of TS-DFM underperform React-OT on different test subsets. The gray, blue, red, and white spheres represent

C, N, O, and H, respectively. Case 1 is a rearrangement reaction in which the reactant converts to its isomer. The large error in TS-DFM originates from inaccurate prediction of dihedral angles. Case 2 is a cleavage reaction where a single molecule breaks into two smaller molecules. TS-DFM predicts the correct chemical bonds but exhibits inaccuracies in predicting the torsional angles and spatial arrangement of the dissociated fragments. Case 3 is a metathesis reaction where two reactant molecules exchange bonds to form two different product molecules. TS-DFM fails to predict the correct bond breaking and also gives inaccurate predictions of torsional angles. Nevertheless, even in some of the worst cases, TS-DFM could still demonstrate partially correct (e.g., correct bond changes) reaction mechanisms.

- defined within a flow matching framework in molecular distance geometry space.
- (2) We describe the detailed design of TSDVNet, responsible for predicting the time-dependent velocity field of TS-DFM. We will introduce the key network module design for mapping the distance geometry of reactant, product and initial guessed TS to the velocity field.
- (3) We provide the mathematical definitions of the metrics and explain their roles in assessing the accuracy of predicted TSs.

- (4) We specify the training configurations for both TS-DFM and MLIP, and list the software packages and hyperparameters for our performed calculations.

Preliminaries of flow matching

Let x_0 and x_1 be data points from a source distribution q_0 and a target distribution q_1 in $\mathbb{R}^d$, respectively. A flow model³⁴ defines a continuous-time Markov process $x_t \in [0, 1]$ via a time-dependent flow ψ_t applied to x_0 , where $x_t = \psi_t(x_0)$ with $x_0 \sim q_0$. The goal of

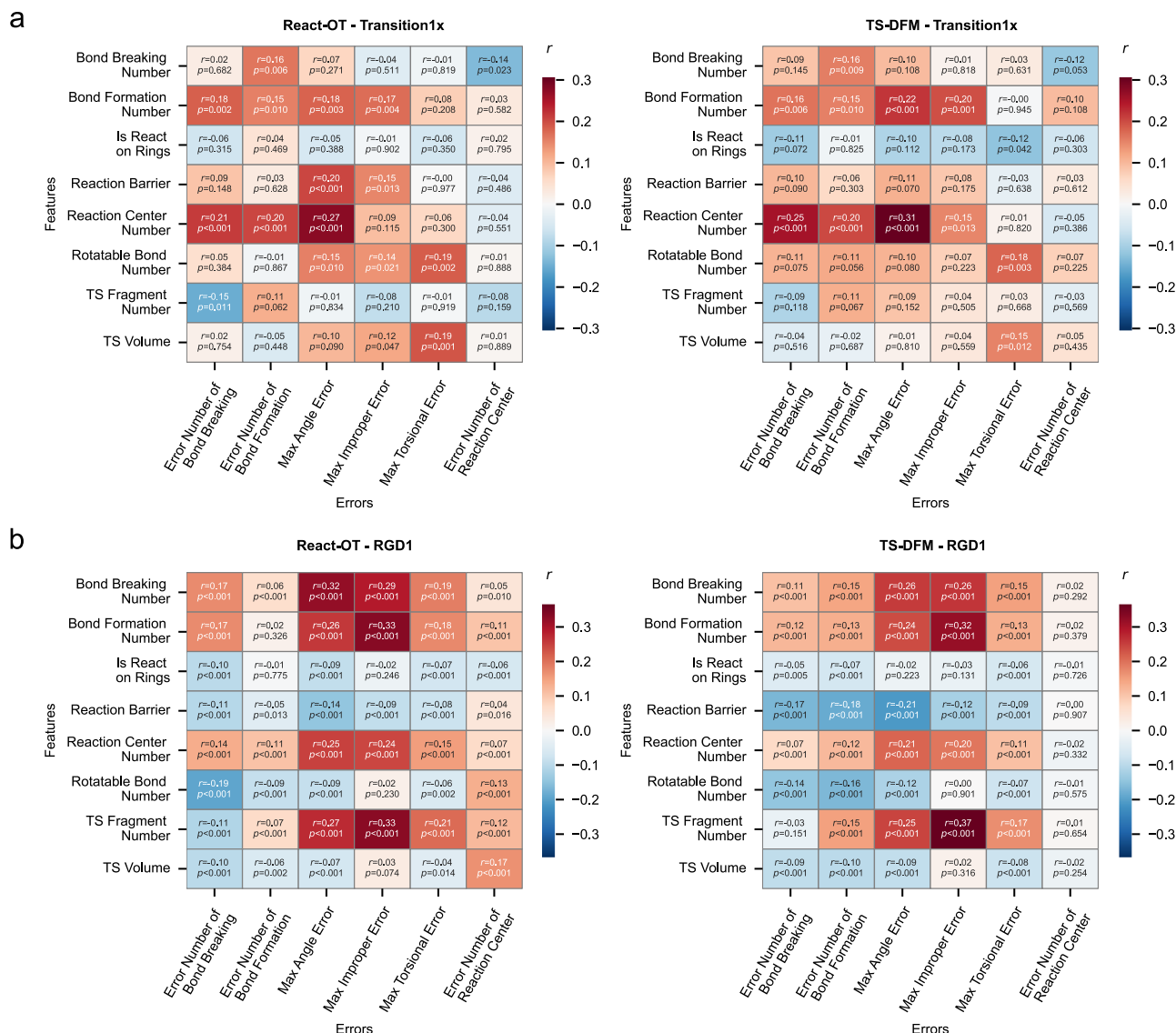


Fig. 5 | Heatmap of the correlation coefficient between eight types of features and six types of prediction errors. a, b Correspond to the correlation coefficient matrices of React-OT and TS-DFM on Transition1x and RGD1 dataset, respectively.

Each grid displays the correlation coefficient r and statistical significance p , with color intensity proportional to the magnitude of r .

generative flow modeling is to learn a flow ψ_t such that $x_1 = \psi_1(x_0)$ follows the target distribution q_1 . The flow ψ_t can be defined by a time-dependent velocity field $u_t: \mathbb{R}^d \rightarrow \mathbb{R}^d$ through an ordinary differential equation (ODE): $\frac{d}{dt}\psi_t(x) = u_t(\psi_t(x))$, with initial condition $\psi_0(x) = x$. A velocity field u_t is said to generate a probability path p_t if its associated flow ψ_t satisfies $x_t = \psi_t(x_0) \sim p_t$ for all $t \in [0, 1]$.

Early work Continuous Normalizing Flow (CNF)³⁵ utilizes black-box ODE solvers to simulate the solution for training a parameterized flow model ψ_t . Due to the computational burden and the gradient vanishing or exploding issues of numerical ODE simulations, CNFs are difficult to train. To address these problems, Flow Matching (FM)³⁶ offers a simulation-free approach by regressing a neural network $u_\theta(x, t)$ to a target velocity field $u_t(x)$. Since $u_t(x)$ is generally intractable, a common practice is to employ the Conditional Flow Matching (CFM) loss^{36,37}, which is guaranteed to yield the same gradients with the FM objective:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t \sim [0, 1], x \sim p_t(x|z), z \sim q(z)} \|\psi_\theta(x, t) - u_t(x|z)\|_2^2, \quad (1)$$

where z denotes any condition independent of x , and $u_t(x|z)$ is a conditional velocity field that generates $p_t(x|z)$ from $p_0(x|z)$.

There exist different ways to define $p_t(x|z)$ and $u_t(x|z)$. Our work is inspired by OT-CFM³⁷ that utilizes the Gaussian conditional optimal transport path to define $p_t(x|z)$ and $u_t(x|z)$. OT-CFM sets

$$z = (x_0, x_1) \text{ and } q(z) = \pi(x_0, x_1), \quad (2)$$

$$u_t(x|z) = x_1 - x_0 \text{ and } p_t(x|z) = \mathcal{N}(x|t \cdot x_1 + (1-t) \cdot x_0, \sigma^2),$$

where π is the optimal coupling between distributions q_0 and q_1 under the squared Euclidean cost, i.e.,

$$\pi = \arg \inf_{\pi \in \Pi} \int c(x_0, x_1) \pi'(dx_0, dx_1), \quad (3)$$

where Π is the set of couplings and $c(x_0, x_1) = \|x_0 - x_1\|_2^2$ is the squared Euclidean distance. $u_t(x|z) = x_1 - x_0$ stems from the optimal transport trajectory under the squared Euclidean cost, which is a straight line with constant velocity. For this choice of z , $p_t(x|z)$ and $u_t(x|z)$, the corresponding marginal distribution $p_t(x)$ has the following boundary

conditions

$$\begin{aligned} p_0(x) &= (q_0(\cdot) * \mathcal{N}(\cdot | 0, \sigma^2))(x), \\ p_1(x) &= (q_1(\cdot) * \mathcal{N}(\cdot | 0, \sigma^2))(x), \end{aligned} \quad (4)$$

where $*$ denotes convolution operator. As $\sigma \rightarrow 0$, the marginal boundary probabilities approach q_0 and q_1 respectively.

Flow matching on distance geometry for TS prediction

The dynamic evolution of interatomic distances between atom pairs is crucial for describing chemical reactions. However, previous state-of-the-art methods such as OA-ReactDiff and React-OT operate in Cartesian coordinate space to predict TSs, which obscures this evolution process. This potentially undermines the quality of their generated TSs. Therefore, our proposed TS-DFM explicitly operates on the space of distance geometry to generate pairwise distance matrix of TS, which improves the accuracy of the predicted TS.

In distance geometry space, the task of double-ended TS prediction can be formulated as generating the pairwise distance matrix of TS $\mathbf{D}_{\text{TS}}$ given the pairwise distance matrices of reactant $\mathbf{D}_{\text{R}}$ and product $\mathbf{D}_{\text{P}}$, along with the atom type vector $\mathbf{Z} = [z_1, \dots, z_n]$, $z_i \in \{1, \dots, 118\}$. Traditional TS optimization algorithms, such as NEB and Hessian-based methods, solve this problem by iteratively refining an initial guess of TS geometry towards the saddle point. From a probabilistic perspective, this process can be abstracted mathematically as transforming a distribution of initial guesses to a target distribution of true TSs.

Formally, let $\mathbb{P}(\mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z})$ denote the probability distribution over chemical reactions, characterized by $\mathbf{D}_{\text{R}}$, $\mathbf{D}_{\text{P}}$, and $\mathbf{Z}$. For any reaction sampled from $\mathbb{P}(\mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z})$, the algorithm operates on a pair of structures: (1) The initial guessed distance geometry of TS, denoted as $\mathbf{D}_{\text{TS},0}$, drawn from a conditional distribution $q_0(\mathbf{D}_{\text{TS},0} | \mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z})$; (2) The true TS distance geometry $\mathbf{D}_{\text{TS},1}$, modeled by a conditional distribution $q_1(\mathbf{D}_{\text{TS},1} | \mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z}, \mathbf{D}_{\text{TS},0}) = \delta(\mathbf{D}_{\text{TS},1} - f(\mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z}, \mathbf{D}_{\text{TS},0}))$, where δ denotes the Dirac delta distribution and f represents the optimization algorithm.

However, the optimization algorithm f requires accurate PES, which relies on computationally prohibitive quantum chemistry calculations. To circumvent this bottleneck, we propose using the Optimal Transport-Conditional Flow Matching (OT-CFM) framework to learn a data-driven surrogate as a substitution for the transport process guided by the PES. In this approach, the distributions of initial guessed TSs and true TSs are treated as the source and target distributions, respectively, and a velocity field is learned to transport samples from the source to the target.

To utilize OT-CFM for generating TS, we first define the optimal coupling π between the source and target distribution. To maintain chemical validity, both $\mathbf{D}_{\text{TS},0}$ and $\mathbf{D}_{\text{TS},1}$ must correspond to the same reaction context $(\mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z})$. Therefore, we define the following joint distribution π for this constraint:

$$\begin{aligned} \pi(\mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z}, \mathbf{D}_{\text{TS},0}, \mathbf{D}_{\text{TS},1}) \\ = \mathbb{P}(\mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z}) \cdot q_0(\mathbf{D}_{\text{TS},0} | \mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z}) \cdot q_1(\mathbf{D}_{\text{TS},1} | \mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z}, \mathbf{D}_{\text{TS},0}). \end{aligned} \quad (5)$$

Under this coupling π , any sample $(\mathbf{D}_{\text{R}}^i, \mathbf{D}_{\text{P}}^i, \mathbf{Z}^i, \mathbf{D}_{\text{TS},0}^i, \mathbf{D}_{\text{TS},1}^i)$ guarantees that $\mathbf{D}_{\text{TS},0}^i$ and $\mathbf{D}_{\text{TS},1}^i$ are paired through the same reaction context $(\mathbf{D}_{\text{R}}^i, \mathbf{D}_{\text{P}}^i, \mathbf{Z}^i)$. Thus, the iterative refinement in traditional algorithms can be viewed as a deterministic transport trajectory from $\mathbf{D}_{\text{TS},0}^i$ to $\mathbf{D}_{\text{TS},1}^i$, guided by the local PES defined by $(\mathbf{D}_{\text{R}}^i, \mathbf{D}_{\text{P}}^i, \mathbf{Z}^i)$.

Then we construct the source and target distributions $q_0(\mathbf{D}_{\text{TS},0} | \mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z})$ and $q_1(\mathbf{D}_{\text{TS},1} | \mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z})$ representing the distributions of initial guessed and true TS over chemical reactions

respectively:

$$\begin{aligned} q_0(\mathbf{D}_{\text{TS},0} | \mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z}) &= \mathbb{P}(\mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z}) \cdot q_0(\mathbf{D}_{\text{TS},0} | \mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z}), \\ q_1(\mathbf{D}_{\text{TS},1} | \mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z}) &= \mathbb{P}(\mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z}) \cdot q_1(\mathbf{D}_{\text{TS},1} | \mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z}). \end{aligned} \quad (6)$$

For the source distribution q_0 , since it is typically chosen as an easy-to-sample distribution and is not required to be Gaussian in OT-CFM³⁷, we adopt a practical choice inspired by the IDPP interpolation method²³:

$$q_0(\mathbf{D}_{\text{TS},0} | \mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z}) = \delta\left(\mathbf{D}_{\text{TS},0} - \frac{\mathbf{D}_{\text{R}} + \mathbf{D}_{\text{P}}}{2}\right). \quad (7)$$

This choice reduces the discrepancy between source and target distribution compared to generally utilized Gaussian noise as source distribution in flow matching model, and provides chemically more plausible initial structures than Cartesian linear interpolation used in React-OT and PSI-based model. For $q_1(\mathbf{D}_{\text{TS},1} | \mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z})$, we assume the following approximation:

$$q_1(\mathbf{D}_{\text{TS},1} | \mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z}) \approx q_1(\mathbf{D}_{\text{TS},1} | \mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z}, \mathbf{D}_{\text{TS},0}), \quad (8)$$

which allows modeling TS generation without explicit dependence on the initial guess $\mathbf{D}_{\text{TS},0}$, though this simplification may not hold in complex reactions where different initial guesses lead to varied TSs.

The coupling π between initial and optimized TSs from the same reaction is used without explicitly solving the optimal transport problem. The time-dependent probability path p_t and velocity field $\mathbf{u}_t$ are derived from the Gaussian probability path minimizing the 2-Wasserstein distance:

$$\begin{aligned} p_t(\mathbf{D}_{\text{TS},t} | \mathbf{Z}, \mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{D}_{\text{TS},1}) &= \mathcal{N}(\mathbf{D}_{\text{TS},t} | t \cdot \mathbf{D}_{\text{TS},1} + (1-t) \cdot \mathbf{D}_{\text{TS},0}, \sigma^2), \\ \mathbf{u}_t(\mathbf{D}_{\text{TS},t} | \mathbf{Z}, \mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{D}_{\text{TS},1}) &= \mathbf{D}_{\text{TS},1} - \mathbf{D}_{\text{TS},0}. \end{aligned} \quad (9)$$

Finally, we can construct a model $\mathbf{v}_{\theta}(\mathbf{D}_{\text{TS},t}, t | \mathbf{Z}, \mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}})$ with learnable parameters θ to approximate the velocity field $\mathbf{u}_t$ by minimizing the loss function:

$$\mathcal{L}_{\text{TS-DFM}}(\theta) = \mathbb{E}_{t, p_t} \|\mathbf{v}_{\theta}(\mathbf{D}_{\text{TS},t}, t | \mathbf{Z}, \mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}) - (\mathbf{D}_{\text{TS},1} - \mathbf{D}_{\text{TS},0})\|_2^2. \quad (10)$$

With a learned velocity field $\mathbf{v}_{\theta}(\mathbf{D}_{\text{TS},t}, t | \mathbf{Z}, \mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}})$, predicted distance geometry of TS $\hat{\mathbf{D}}_{\text{TS},1}$ is obtained by numerical ODE simulation starting from $\mathbf{D}_{\text{TS},0}$. The corresponding Cartesian coordinates of predicted TS $\hat{\mathbf{R}}_{\text{TS}}$ is recovered by nonlinear optimization, which will be elaborated later. We also provide pseudocodes for the training and inference of TS-DFM in Suppl. Alg. 2 and 3 respectively.

Network structure of TSDVNet

The detailed network structure TSDVNet for constructing $\mathbf{v}_{\theta}(\mathbf{D}_{\text{TS},t}, t | \mathbf{Z}, \mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}})$ in the flow matching framework is illustrated in Fig. 1b. The model takes atom type $\mathbf{Z}$, timestep t , and three pairwise distance matrices $\mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{D}_{\text{TS},t}$ as input and then outputs the predicted velocity field. As shown in Fig. 1b, the network contains two branches with the same architecture but different parameters. The upper branch aims to generate the representations of $\mathbf{Z}$ and $\mathbf{D}_{\text{TS},t}$, which are then used to predict the velocity field. While the lower branch is used to generate the conditional representations, where the conditional inputs $\mathbf{D}_{\text{R}}, \mathbf{D}_{\text{P}}, \mathbf{Z}$ corresponding to the reactant and product will be processed. To further enhance the learned representations and combining the information of reactant, product and TS, we also employ feature fusion operations between the two branches.

Within each branch, both atom and pair representations will be computed. The Atom Embedding and Pair Embedding blocks encode atom type $\mathbf{Z}$, pairwise distance matrix $\mathbf{D}$ and timestep t into atom representations $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{R}^{n \times d_x}$ and pair representations $\mathbf{P} = [\mathbf{p}_{ij}]_{i,j=1}^n \in \mathbb{R}^{n \times n \times d_p}$, where d_x and d_p represents the dimension of

atom and pair representations respectively. Then, these representations will be processed by L update blocks. In the l -th update block, the atom representations will be first updated by Cross-Molecule Message Passing block, fusing the information of $\mathbf{x}_R^{(l)}$, $\mathbf{x}_P^{(l)}$ and $\mathbf{x}_{TS,t}^{(l)}$. Then, the Inner-Molecule Message Passing block updates the atom representations using the neighboring information within each molecule. Utilizing the updated atom representations, the pair features will be updated by Mixing Update and Triangular Update blocks, focusing on local and global pair information respectively. Finally, the Pairwise Distance Flow Prediction block utilizes the atom and pair representations $\mathbf{x}_{TS,t}$, $\mathbf{p}_{TS,t}$ to predict the velocity field $\mathbf{v}$. In the following content of this section, each module will be introduced in detail.

The Atom Embedding block encodes the atom type $\mathbf{Z}$, the neighboring information $j \in N_{\text{neigh}}(i)$, and flow timestep t into feature vectors $\mathbf{X}$, which is defined as

$$\begin{aligned} \mathbf{x}_i &= \text{Embedding}(\mathbf{Z}_i); \\ \mathbf{x}_i &\leftarrow \mathbf{x}_i + \text{Time_Embedding}(t); \\ \mathbf{x}_i &\leftarrow \mathbf{W}_N \left[\mathbf{x}_i, \sum_{j \in N_{\text{neigh}}(i)} \text{Embedding}(\mathbf{x}_j) \odot (\mathbf{W}_F \cdot \text{RBF}(d_{ij})) \right] + \mathbf{b}_N, \end{aligned} \quad (11)$$

where [...] denotes concatenation operation. Unless stated otherwise, $\mathbf{W}_{\square}$ and $\mathbf{b}_{\square}$ are used to represent learnable parameters. Within this block, the Embedding function maps the atom types into embedded vectors, and the Time_EMBEDDING function encodes the timestep t into a vector by

$$\begin{aligned} \mathbf{m}_t &= t * \exp\left(-\frac{\log(10000)}{d/2 - 1} * [0, 1, \dots, \frac{d}{2} - 1]\right); \\ \text{output} &= [\sin(\mathbf{m}_t), \cos(\mathbf{m}_t)], \end{aligned} \quad (12)$$

and the RBF function encodes the atomic distances d_{ij} into vectors with

$$\mathbf{e}_{\text{RBF}}(d_{ij}) = \phi(d_{ij}) \exp(-\beta_k(\exp(-d_{ij}) - \mu_k)^2), \quad (13)$$

where

$$\phi(d_{ij}) = \begin{cases} \frac{1}{2} \left(\cos\left(\frac{\pi d_{ij}}{d_{\text{cut}}}\right) + 1 \right), & \text{if } d_{ij} \leq d_{\text{cut}} \\ 0, & \text{if } d_{ij} > d_{\text{cut}} \end{cases}, \quad (14)$$

β_k and μ_k are trainable parameters initialized by the settings in ref. 38, and d_{cut} is a hyperparameter representing the cutoff distance. Based on the embedded atom features $\mathbf{X}$, the Edge Embedding block generates pair representations $\mathbf{P}$ by

$$\mathbf{p}_{ij} = \mathbf{W}_E [\mathbf{W}_S \mathbf{x}_i + \mathbf{b}_S, \mathbf{W}_D \mathbf{x}_j + \mathbf{b}_D, \mathbf{W}_r \mathbf{e}_{\text{RBF}}(d_{ij}) + \mathbf{b}_r] + \mathbf{b}_E. \quad (15)$$

At the l -th update block, the Cross Molecule Message Passing block and Inner Molecule Message Passing block are used to update atom representations. Taking atom representation $\mathbf{x}_{i,TS,t}$ (the i -th atom representation in $\mathbf{x}_{TS,t}$) as example, the Cross-Molecule Message Passing block in the upper branch of the network shown in Fig. 1b updates $\mathbf{x}_{i,TS,t}$ by fusing the cross-molecule information as follows

$$\begin{aligned} \mathbf{x}_{i,TS,t} &\leftarrow \text{LayerNorm} \left(\mathbf{x}_{i,TS,t} + \sum_{j \in N_{\text{neigh}}(i,TS)} \text{Attn}(\mathbf{x}_{i,R}, \mathbf{x}_j, \mathbf{p}_{ij,TS,t}, \mathbf{p}_{ij,TS,t}, d_{ij,TS,t}) \right. \\ &\quad \left. + \sum_{j \in N_{\text{neigh}}(i,TS)} \text{Attn}(\mathbf{x}_{i,P}, \mathbf{x}_j, \mathbf{p}_{ij,TS,t}, \mathbf{p}_{ij,TS,t}, d_{ij,TS,t}) \right), \end{aligned} \quad (16)$$

where $j \in N_{\text{neigh}}(i,TS)$ denotes the neighbors of i -th atom determined by $\mathbf{D}_{TS,t}$. The atom representations $\mathbf{x}_{i,R}$ and $\mathbf{x}_{i,P}$ are updated in the same

manner, using the corresponding block in the lower branch. Then, the Inner-Molecule Message Passing blocks update atom representation $\mathbf{x}_i$ by utilizing the same Attn function to implement inner-molecule message passing as

$$\mathbf{x}_i \leftarrow \text{LayerNorm} \left(\mathbf{x}_i + \sum_{j \in N_{\text{neigh}}(i)} \text{Attn}(\mathbf{x}_i, \mathbf{x}_j, \mathbf{p}_{ij}, d_{ij}) \right). \quad (17)$$

The Attn function is used to compute interactions between pairs of atoms through the modified attention mechanism. Take $\text{Attn}(\mathbf{x}_i, \mathbf{x}_j, \mathbf{p}_{ij}, d_{ij})$ as example, its output is calculated by

$$\begin{aligned} \mathbf{Q} &= \mathbf{W}_Q \mathbf{x}_i + \mathbf{b}_Q; \\ \mathbf{K} &= \mathbf{W}_K \mathbf{x}_j + \mathbf{b}_K; \\ \mathbf{D}_K &= \mathbf{W}_{D_K} \mathbf{e}_{\text{RBF}}(d_{ij}) + \mathbf{b}_{D_K}; \\ \mathbf{V} &= \mathbf{W}_V \mathbf{x}_j + \mathbf{b}_V; \\ \mathbf{D}_V &= \mathbf{W}_{D_V} \mathbf{e}_{\text{RBF}}(d_{ij}) + \mathbf{b}_{D_V}; \\ \mathbf{D}_P &= \mathbf{W}_P \mathbf{p}_{ij} + \mathbf{b}_P; \\ \text{output} &= \mathbf{V} \odot \mathbf{D}_V \odot \text{SiLU}(\mathbf{Q} \odot \mathbf{K} \odot \mathbf{D}_K + \mathbf{D}_P) \cdot \phi(d_{ij}), \end{aligned} \quad (18)$$

where $\odot$ represents the Hadamard product of two matrices.

After updating the atom representations, the Edge Update module first updates the pair representation $\mathbf{p}_{ij}$ using its connected atom representations $\mathbf{x}_i, \mathbf{x}_j$, which is defined as

$$\begin{aligned} \mathbf{x}_{\text{map}} &= \mathbf{W}_{\text{map}} [\mathbf{x}_i, \mathbf{x}_j] + \mathbf{b}_{\text{map}}; \\ \mathbf{p}_a &= \mathbf{W}_a [\mathbf{x}_{\text{map}}, \mathbf{p}_{ij}] + \mathbf{b}_a; \\ \mathbf{p}_b &= \mathbf{W}_b [\mathbf{x}_{\text{map}}, \mathbf{p}_{ij}] + \mathbf{b}_b; \\ \Delta \mathbf{p}_{ij} &= \mathbf{W}_{ab} \cdot \text{Flatten}(\mathbf{p}_a \otimes \mathbf{p}_b) + \mathbf{b}_{ab}; \\ \mathbf{p}_{ij} &\leftarrow \text{LayerNorm}(\mathbf{p}_{ij} + \Delta \mathbf{p}_{ij}), \end{aligned} \quad (19)$$

where $\otimes$ denotes the Kronecker product of two vectors. After that, the Edge Triangular Update aggregates the information that forms triangular connections with atom i, j to further enhance the pair representations, which is adopted from the implementation in AlphaFold2³⁹ and computed by

$$\begin{aligned} \mathbf{p}_{ij,m} &= \text{Sigmoid}(\mathbf{W}_{m_1} \mathbf{p}_{ij} + \mathbf{b}_{m_1}) \odot (\mathbf{W}_{m_2} \mathbf{p}_{ij} + \mathbf{b}_{m_2}); \\ \mathbf{p}_{ij,n} &= \text{Sigmoid}(\mathbf{W}_{n_1} \mathbf{p}_{ij} + \mathbf{b}_{n_1}) \odot (\mathbf{W}_{n_2} \mathbf{p}_{ij} + \mathbf{b}_{n_2}); \\ \mathbf{o}_{ij} &= \text{LayerNorm} \left(\sum_k \mathbf{p}_{ik,m} \odot \mathbf{p}_{jk,n} + \sum_k \mathbf{p}_{ki,m} \odot \mathbf{p}_{kj,n} \right); \\ \mathbf{p}_{ij} &\leftarrow \text{LayerNorm}(\mathbf{p}_{ij} + \text{Sigmoid}(\mathbf{W}_{o_1} \mathbf{p}_{ij} + \mathbf{b}_{o_1}) \odot (\mathbf{W}_{o_2} \mathbf{o}_{ij} + \mathbf{b}_{o_2})). \end{aligned} \quad (20)$$

Finally, the Pairwise Distance Flow Prediction module predicts the velocity field $\mathbf{v}_{TS,t}$ by incorporating the atom and pair representations after L update blocks, which is calculated by

$$\begin{aligned} \mathbf{x}_{\text{map}} &= \mathbf{W}_{\text{map}} [\mathbf{x}_i, \mathbf{x}_j] + \mathbf{b}_{\text{map}}; \\ \mathbf{p}_a &= \mathbf{W}_a [\mathbf{x}_{\text{map}}, \mathbf{p}_{ij}] + \mathbf{b}_a; \\ \mathbf{p}_b &= \mathbf{W}_b [\mathbf{x}_{\text{map}}, \mathbf{p}_{ij}] + \mathbf{b}_b; \\ \mathbf{v}_{TS,t} &= \mathbf{W}_{ab} \cdot \text{Flatten}(\mathbf{p}_a \otimes \mathbf{p}_b) + \mathbf{b}_{ab} \end{aligned} \quad (21)$$

With the aforementioned network design, the velocity field $\mathbf{v}_{TS,t}$ exhibits permutation equivariance. More importantly, the dependence on reaction direction, i.e., the exchange of reactant and product, will also be eliminated.

Reconstruct molecular structure from distance geometry

In TS-DFM, a velocity field of molecular distance geometry is learned for generating the pairwise distance matrix of TS. Given atomic coordinates $\vec{\mathbf{R}} = [\mathbf{r}_1, \dots, \mathbf{r}_n] \in \mathbb{R}^{n \times 3}$, the pairwise distance matrix $\mathbf{D}$ can be directly calculated from $\vec{\mathbf{R}}$, with elements $d_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\|_2$. However, reconstructing the coordinates $\vec{\mathbf{R}}$ from a distance matrix $\mathbf{D}$ is a non-trivial inverse problem that requires nonlinear optimization.

Starting from an initial atomic coordinates $\vec{\mathbf{R}}_0$, the optimal coordinates $\vec{\mathbf{R}}^*$ are obtained by minimizing the weighted discrepancy between the predicted distances $\hat{d}_{ij}$ and the distances in the reconstructed geometry:

$$\vec{\mathbf{R}}^* = \arg \min_{\vec{\mathbf{R}}} \sum_{ij} \omega_{ij} (\hat{d}_{ij} - \|\mathbf{r}_i - \mathbf{r}_j\|_2)^2. \quad (22)$$

where ω_{ij} denotes the weighting coefficient. We evaluate three weighting strategies (see Supplementary Tables 7, 8), all of which yield similar reconstruction accuracy. To emphasize the molecular bonding structure and reduce the influence of larger errors in long-range distance predictions, we adopt $\omega_{ij} = 1/\hat{d}_{ij}^2$ as default. The optimization is performed using the L-BFGS algorithm.

A known limitation of distance geometry lies in its inability to distinguish enantiomers, since they share identical pairwise distances. In our study, however, the three-dimensional structures of both the reactant and the product are known. These configurations define the specific stereochemistry of the starting and ending points of the reaction path. Therefore, the TS we aim to locate is not an arbitrary saddle point, but the specific one that connects these two structures. To incorporate this stereochemical information, we generate the initial structure for nonlinear optimization $\vec{\mathbf{R}}_0$ via linear interpolation between the reactant and product geometries. This inherently transfers the stereochemistry from the endpoints into the initial TS structure. Consequently, the subsequent nonlinear optimization acts as a conditional refinement anchored to the reactant and product structures, thereby preserving the stereochemical configuration throughout the reconstruction process. We validate this approach using the Transition1x test set. When reconstructing TS geometries from their exact distance matrices, we achieve a maximum RMSD of 0.0031 Å. Such low RMSD values generally correspond to accurate spatial arrangements, indicating high-fidelity TS geometry recovery. The complete reconstruction procedure is summarized in Suppl. Alg. 1.

Evaluation metrics

As mentioned above, the TSs are located at the saddle points on the PES. In evaluating the quality of predicted TSs, both their structural similarity to the reference TS and the properties associated with saddle points should be considered. Therefore, the evaluation metrics used in our work are classified into two categories. One category aims to evaluate the structural accuracy, including the following metrics:

- (1) Root Mean Squared Distance (RMSD):

$$\text{RMSD} = \sqrt{\frac{1}{n} \sum_{i=1}^n \|\hat{\mathbf{r}}_i - \mathbf{r}_i\|_2^2}, \quad (23)$$

where $\hat{\mathbf{r}}_i$ and $\mathbf{r}_i$ denote the i -th atomic coordinates of predicted and true TS respectively, and $\hat{\mathbf{r}}_i$ is aligned w.r.t. $\mathbf{r}_i$ by Kabsch alignment algorithm⁴⁰ before calculating RMSD. It mainly represents the global structural accuracy of the predicted TS.

- (2) Distance Mean Absolute Error (DMAE):

$$\text{DMAE} = \frac{1}{n(n-1)} \sum_{i \neq j} |\hat{d}_{ij} - d_{ij}|, \quad (24)$$

where $d_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\|_2$ and $\hat{d}_{ij} = \|\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j\|_2$. Compared with RMSD, DMAE focuses on local structural accuracy, making it a complementary metric for more nuanced structural accuracy evaluation.

Another group of metrics are related to the potential energy properties of the predicted TS, which are derived from DFT calculations:

- (1) Potential energy error of TS $|\Delta E_{\text{TS}}|$:

$$|\Delta E_{\text{TS}}| = |\hat{E}_{\text{TS}} - E_{\text{TS}}|, \quad (25)$$

where $\hat{E}_{\text{TS}}$ and E_{TS} denote the potential energy of the predicted and true TS. As the cases shown in ref. 41, a predicted TS with large $|\Delta E_{\text{TS}}|$ often exhibits an imperfect geometric structure that has not fully converged to saddle point, even with a low RMSD. Therefore, small $|\Delta E_{\text{TS}}|$ is a fundamental requirement for a valid result.

- (2) The root mean square force norm of predicted TS F_{rms} :

$$F_{\text{rms}} = \sqrt{\frac{1}{n} \sum_{i=1}^n \|\hat{\mathbf{f}}_i\|_2^2}, \quad (26)$$

where $\hat{\mathbf{f}}_i = \nabla_{\mathbf{r}_i} \hat{E}_{\text{TS}}$. A low F_{rms} could prove that the structure has converged to a stationary point. It serves as a prerequisite for further frequency analysis.

- (3) The maximum force norm of the predicted TS f_{max} :

$$f_{\text{max}} = \max_{i=1, \dots, n} \|\hat{\mathbf{f}}_i\|_2. \quad (27)$$

Whereas F_{rms} measures global convergence, f_{max} identifies localized geometric strain in a predicted TS, thus providing non-redundant information that supplements F_{rms} .

- (4) % of saddle points: The percentage of predicted TSs in the test set that have exactly one negative frequency—a key characteristic of saddle point. These frequencies are derived from harmonic analysis of the Hessian matrix. It directly indicates whether the predicted TS is located at the saddle point of PES or not.

To summarize, the evaluation metrics described above assess the quality of predicted TSs from both structural and energetic perspectives. In comparison with previous works^{20–22}, our study employs a more detailed set of metrics, thereby enabling a more comprehensive evaluation of different TS prediction methods.

Details of model training

The learnable velocity field network $\mathbf{v}_\theta$ is based on our proposed TSDVNet. This network comprises 6 update blocks, where both the atom and pair representations have a dimension of 128. The cutoff distance is set as 20 Å. We use mini-batch gradient descent algorithm with batch size of 32 to minimize $\mathcal{L}_{\text{TS-DFM}}$ using Adam optimizer⁴², whose learning rate is initialized as $5e-4$ and will be decayed by 0.8 if the performance is not improved for more than 40 epochs. The noise scale σ is set as 0.1 to improve the robustness of the learned velocity field. During inference, the default timestep of numerical ODE simulation is set to 0.05. Ablation studies regarding the choice of these hyperparameters are in Supplementary Tables 3, 4.

In our experiments on Transition1x dataset, a MLIP $E_\phi(\mathbf{Z}, \vec{\mathbf{R}})$ with learnable parameters ϕ is trained and then utilized as a surrogate PES for various tasks. We utilize TorchMD-NET⁴³, an efficient and powerful equivariant graph neural network structure, as the backbone of the MLIP. The number of message passing layers is set as 6, the hidden

dimension is set as 256, and the cutoff distance is set as 10 Å. Φ are learned by optimizing the following loss function

$$\mathcal{L}_{\text{pot}} = \lambda(E_{\Phi} - E)^2 + \frac{1 - \lambda}{3n} \|\hat{\mathbf{F}} - \mathbf{F}\|^2, \quad (28)$$

where E_{Φ} and E denotes the predicted and true potential energy respectively, $\hat{\mathbf{F}} = -\nabla_{\mathbf{R}} E_{\Phi}$ denotes the predicted atomic forces, and $\mathbf{F}$ represents the true atomic forces. All the molecular configurations in the training and validation set are used for model training and validation, respectively. We use mini-batch gradient descent algorithm with batch size of 50 to minimize the loss function using Adam optimizer⁴². The learning rate is initialized as 1e-3 and will be decayed by 0.75 if the performance is not improved for more than 25 epochs. The weight factor λ in the loss function is set as 0.01 to put more emphasis on atomic forces, which is crucial for NEB calculation. The trained model achieves Mean Absolute Error (MAE) of 0.0696 eV for energy prediction and 0.0712 eV/Å for atomic force prediction on the test set.

Calculation details

In this work, the NEB calculation is implemented by utilizing the interfaces of the Atomic Simulation Environment (ASE)⁴⁴. NEB is a method for finding the MEP and TS given the reactant and product of a chemical reaction. It begins by optimizing the geometries of the reactant and product to ensure they represent local minima on the PES. The reactants and products are relaxed for at most 500 steps using the BFGS optimizer with $\alpha = 70$ and a maximum step size of 0.005 Å. They are considered relaxed if the norm of the forces is lower than 0.01 eV/Å. A series of intermediate configurations are then created via interpolation between two endpoints, forming a chain connected by virtual spring forces that maintain uniform spacing along the path. We use eleven images to represent the MEP, whose initial configurations are generated by the IDPP²³ interpolation method. If a predicted TS is provided, eight intermediate images are first placed along the reactant-TS and TS-product segments in a ratio determined by their respective DMAE values. Each segment is then initialized using IDPP interpolation method, with reactant, TS, and product structures fixed as anchor points to constrain the initialization process. During optimization, forces acting on each image are decomposed: the parallel component is removed to prevent sliding, while the perpendicular nudging force drives images toward the MEP. Since the highest-energy image may not be the true saddle point, CI-NEB further maximizes energy along the path. Our NEB calculations are performed for at most 1000 steps, where the spring constant is set to 0.1 and the method is chosen as `aseneb`. The MEP is found by the optimizer proposed in ref. 45 with the default parameter. During iteration, the path is considered converged if the maximal perpendicular force on the path is below 0.05 eV/Å.

The Hessian-based saddle point optimization are performed by utilizing Sella¹². Sella provides an automated algorithm that optimizes molecular structures to saddle points on the PES. It could automatically generate internal coordinates from Cartesian geometries, with the ability to handle problematic coordinates during iteration. Instead of evaluating the full Hessian, it uses iterative diagonalization and TS-BFGS updates. For saddle point optimization, it combines null-space sequential quadratic programming (SQP) with restricted-step partitioned rational function optimization (RS-PRFO) to move toward saddle points by maximizing energy along the reaction coordinate while minimizing in other directions. Additionally, it employs geodesic stepping to realize displacements in redundant internal coordinates, accounting for the curvature of the coordinate manifold to ensure physically valid configurations and improve convergence efficiency. In our implementations, we utilize the default parameters provided by

Sella. The optimization runs for at most 1000 steps with a convergence threshold of 0.05 eV/Å.

The intrinsic reaction coordinate (IRC) calculation aims to trace the MEP on the PES, connecting a TS back to its corresponding reactant and product. Starting precisely from the located TS geometry, the IRC path follows the direction of steepest descent in mass-weighted coordinates downhill towards lower energy. This involves numerical integration of the differential equations governing the path, typically using small steps along the negative gradient of the PES, while accounting for atomic masses to define the intrinsic reaction coordinate. The calculation proceeds iteratively in forward and reverse directions from the TS until it reaches stable minima corresponding to the reactant and product states, thereby revealing the complete reaction mechanism. We utilize the `IRC` method provided in Sella for IRC calculations, where the parameters are set as default. The calculation runs for a maximum of 1000 steps with a convergence threshold of 0.1 eV/Å. Then, the reactant and product structures will be further relaxed to ensure they are located at the local energy minima, following the same relaxation operations utilized in NEB calculations.

Normal mode sampling generates molecular configurations by exploiting the harmonic approximation around a structure located at the local minimum of PES²⁸. It first calculates the system's vibrational normal modes, i.e., the eigenvectors of the mass-weighted Hessian matrix, then randomly displaces atoms along these modes. The amplitudes of random displacements are scaled by the Boltzmann distribution at a specified temperature, ensuring the sampled configurations reflect the correct quantum or classical thermal fluctuations for that temperature. In our implementations, the normal modes are obtained by DFT calculation, and the temperature is set as 300 K to control the amplitude of perturbations.

All DFT calculations are performed using PySCF⁴⁶. In performing calculations, the exchange-correlation functional and basis sets are kept consistent with the dataset. We set the energy tolerance as 1e-6 in atomic units and the maximum number of SCF cycles as 200.

Data availability

The original TransitionIx dataset used in this study are available at Figshare under accession code <https://doi.org/10.6084/m9.figshare.19614657.v4>. The full RGD1 dataset used in this study are available at Figshare under accession code 10.6084/m9.figshare.21066901.v6. The processed RGD1 subset and trained model parameters generated in this study have been deposited at Zenodo under accession code 10.5281/zenodo.20179684⁴⁷. The source data of this work have been deposited at Zenodo under accession code 10.5281/zenodo.20301872⁴⁸.

Code availability

The code developed for this study is available at <https://github.com/Yufei-Luo/TS-DFM>. The archived release is available on Zenodo⁴⁹.

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Author contributions

Y.L.: Conceptualization, methodology, software, writing of original draft, and visualization. X.G.: Methodology, review, editing, and funding. J.S.: Methodology, review, editing, and funding.

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Competing interests

The authors declare no competing interests.

Additional information

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